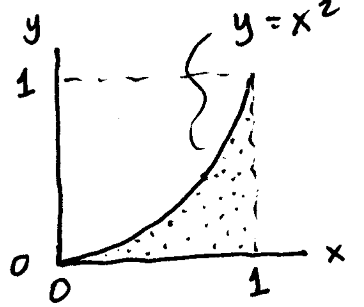
Centroids by Integration

Hibbeler Ex Prob 9.5 worked $\underline{\underline{4}}$ ways



Determine the centroid $(\bar{x}, \bar{y})$ for this area (shaded).

Curve: $y = x^2$

or $x = \sqrt{y}$

Four methods!

① Double Integral.

y limits from $0 \to x^2$

$dA = dy\,dx$

$\tilde{x} = x, \quad \tilde{y} = y$

e.g. $\bar{x}$ numerator: $\int_A x\,dA = \int_0^4 \int_0^{x^2} x\,\underline{\underline{dy}}\,dx$

② Double Integral!

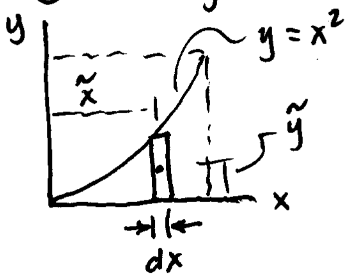
x limits $\sqrt{y}$ to 1

$dA = dx\,dy$

$\tilde{x} = x, \quad \tilde{y} = y$

e.g. $\bar{x}$ numerator: $\int_A x\,dA = \int_0^4 \int_{\sqrt{y}}^1 x\,\underline{\underline{dx}}\,dy$

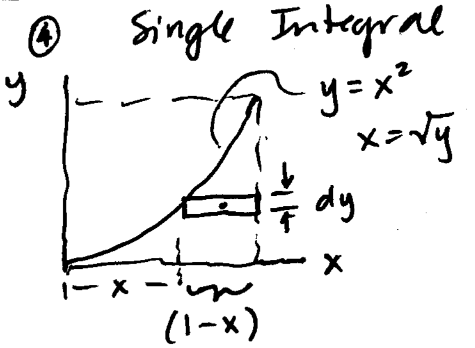
③ Single Integral (Vertical Element):



$dA = y\,dx = x^2\,dx$

$\tilde{x} = x \; ; \quad \tilde{y} = \frac{y}{2}$

④ Single Integral (Horiz Element)



$dA = (1-x)\,dy = (1-\sqrt{y})\,dy$

$\tilde{y} = y, \quad \tilde{x} = x + \frac{1-x}{2}$

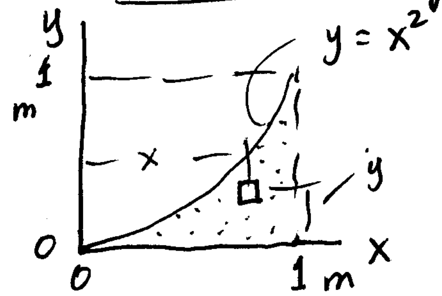
$= x + \frac{1}{2} - \frac{x}{2}$

$\tilde{x} = \frac{x+1}{2}$

Ex Prob 9-5 (cont'd)

① Double Integral, $\boxed{dA = dy\,dx}$

dy limits: $0 \to x^2$
dx limits: $0 \to 1$



$\tilde{x} = x, \quad \tilde{y} = y$

$$\boxed{\bar{x} = \frac{\int_A \tilde{x}\,dA}{\int_A dA}}$$

$$\boxed{\bar{y} = \frac{\int_A \tilde{y}\,dA}{\int_A dA}}$$

Area $A = \int_0^1 \int_0^{x^2} dy\,dx = \int_0^1 x^2\,dx$

$= \frac{1}{3}x^3 \Big|_0^1 \quad \boxed{A = \frac{1}{3} = 0.333 \ m^2}$

TI-89 $\int(\int(1, y, 0, x\hat{\ }2), x, 0, 1) = 0.333 \ m^2$

$\bar{x}$ Numerator: $= \int_0^1 \int_0^{x^2} x\,dy\,dx$

$= \int_0^1 x^3\,dx = \frac{1}{4}x^4 \Big|_0^1 = \frac{1}{4} = 0.25 \ m^3$

TI-89 $\int(\int(x, y, 0, x\hat{\ }2), x, 0, 1) = 0.25 \ m^3$

$\bar{x}$ Centroid $\bar{x} = \frac{0.25}{.333} = \frac{1/4}{1/3} = 3/4 = \boxed{0.75 \ m = \bar{x}}$

$\bar{y}$ Numerator: $= \int_0^1 \int_0^{x^2} y\,dy\,dx = \int_0^1 \frac{1}{2}y^2 \Big|_0^{x^2} dx$

$= \int_0^1 \frac{1}{2}(x^4)\,dx = \frac{1}{10}x^5 \Big|_0^1 = \frac{1}{10} = 0.10 \ m^3$

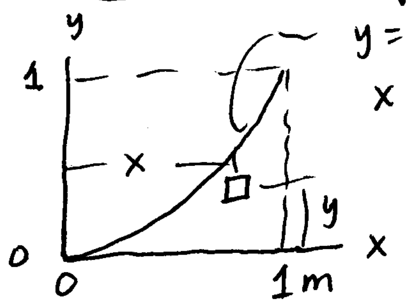
TI-89 $\int(\int(y, y, 0, x\hat{\ }2), x, 0, 1) = 0.10 \ m^3$

$\bar{y}$ Centroid: $\bar{y} = \frac{.1}{.333} = \frac{1/10}{1/3} = \frac{3}{10} = \boxed{0.30 \ m = \bar{y}}$

Ex. Prob 9-5 (cont'd)

② Double Integral, $\boxed{dA = dx\,dy}$ dx limits: $\sqrt{y} \to 1$

dy limits: $0 \to 1$



$y = x^2$,

$x = \sqrt{y}$

$\tilde{x} = x, \quad \tilde{y} = y$

$$\text{Area } A = \int_0^1 \int_{\sqrt{y}}^1 dx\,dy = \int_0^1 (1 - \underbrace{\sqrt{y}}_{y^{1/2}})\,dy$$

$$= \left(y - \tfrac{2}{3} y^{3/2}\right)\Big|_0^1 = 1 - \tfrac{2}{3} = \boxed{\tfrac{1}{3} = .333\ m^2}$$

TI-89: $\int(\int(1, x, y^{\wedge}.5, 1), y, 0, 1) = 0.333\ m^2$

$\bar{x}$ numerator:

$$= \int_0^1 \int_{\sqrt{y}}^1 x\,dx\,dy = \int_0^1 \tfrac{1}{2}x^2\Big|_{\sqrt{y}}^1 dy$$

$$= \int_0^1 \tfrac{1}{2}(1-y)\,dy = \tfrac{1}{2}\left(y - \tfrac{1}{2}y^2\right)\Big|_0^1 = \tfrac{1}{2}\left(1 - \tfrac{1}{2}\right) = \boxed{\tfrac{1}{4}\ m^3}$$

TI-89: $\int(\int(x, x, y^{\wedge}.5, 1), y, 0, 1) = \boxed{.25\ m^3}$

$\bar{x}$ Centroid: $\bar{x} = \dfrac{.25}{.333} = \dfrac{1/4}{1/3} = \boxed{\tfrac{3}{4} = 0.75\ m = \bar{x}}$

$\bar{y}$ Numerator:

$$= \int_0^1 \int_{\sqrt{y}}^1 y\,dx\,dy = \int_0^1 y(1 - \underbrace{\sqrt{y}}_{y^{1/2}})\,dy$$

$$= \int_0^1 (y - y^{3/2})\,dy = \left(\tfrac{1}{2}y^2 - \tfrac{2}{5}y^{5/2}\right)\Big|_0^1 = \tfrac{1}{2} - \tfrac{2}{5} =$$

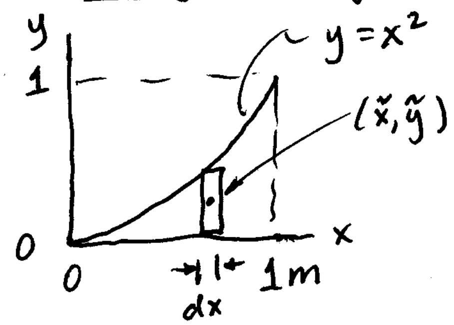
$$= \tfrac{5}{10} - \tfrac{4}{10} = \boxed{\tfrac{1}{10} = 0.10\ m^3}$$

TI-89: $\int(\int(y, x, y^{\wedge}.5, 1), y, 0, 1) = 0.10\ m^3$

$\bar{y}$ Centroid $= \dfrac{.10}{.333} = \dfrac{1/10}{1/3} = \boxed{\tfrac{3}{10} = 0.30\ m = \bar{y}}$

Ex. Prob 9-5 (cont'd)

③ Single Integral, Vertical Element



Since $y=x^2$

$$dA = y\,dx = x^2\,dx$$

$$\tilde{x} = x, \quad \tilde{y} = y/2 = \frac{x^2}{2}$$

$$\text{Area } A = \int_0^1 y\,dx = \int_0^1 x^2\,dx$$

$$= \frac{1}{3}x^3\Big|_0^1 = \boxed{\frac{1}{3} = 0.333\ m^2 = A}$$

TI-89: $\int(x\hat{}2, x, 0, 1) = \boxed{0.333\ m^2}$

$\bar{x}$ Numerator: $= \int_0^1 x\,(\underbrace{x^2 dx}_{dA}) = \int_0^1 x^3\,dx$

$$= \frac{1}{4}x^4\Big|_0^1 = \boxed{\frac{1}{4} = 0.25\ m^3}$$

TI-89: $\int(x\hat{}3, x, 0, 1) = 0.25\ m^3$

$\bar{x}$ Centroid: $\bar{x} = \frac{.25}{.333} = \frac{1/4}{1/3} = \boxed{\frac{3}{4} = 0.75\ m = \bar{x}}$

$\bar{y}$ Numerator: $= \int_0^1 \underbrace{\left(\frac{x^2}{2}\right)}_{"\tilde{y}}\underbrace{(x^2 dx)}_{"dA} = \int_0^1 \frac{1}{2}x^4\,dx$

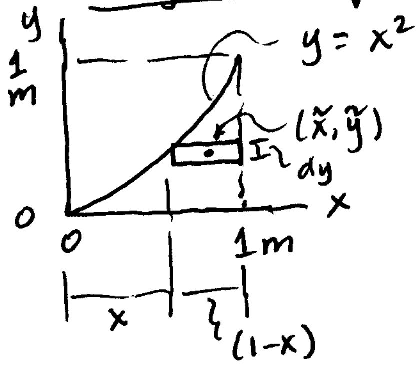
$$= \frac{1}{10}x^5\Big|_0^1 = \boxed{\frac{1}{10} = 0.10\ m^3}$$

TI-89: $= \int(.5 * x\hat{}4, x, 0, 1) = 0.10\ m^3$

$\bar{y}$ centroid: $\bar{y} = \frac{.10}{.333} = \frac{1/10}{1/3} = \boxed{\frac{3}{10} = 0.30\ m = \bar{y}}$

Ex. Prob 9-5 (cont'd)

④ Single Integral, Horiz Element



But $x = \sqrt{y}$

$$dA = (1 - x)\,dy = (1 - \sqrt{y})\,dy$$

$$\tilde{y} = y$$

$$\tilde{x} = x + \frac{1-x}{2} = x + \frac{1}{2} - \frac{x}{2}$$

$$= \frac{x}{2} + \frac{1}{2} \Rightarrow \boxed{\tilde{x} = \frac{x+1}{2} \qquad \tilde{x} = \frac{\sqrt{y}+1}{2}}$$

Area: $A = \int_0^1 (1 - y^{1/2})\,dy = \left(y - \frac{2}{3}y^{3/2}\right)_0^1$

$$A = 1 - \frac{2}{3} = \boxed{\frac{1}{3} = 0.333\ m^2 = A}$$

TI-89: $\int(1 - y\text{^}.5, y, 0, 1) = \boxed{0.333\ m^2}$

$\bar{x}$ **Numerator:** $= \int_0^1 \underbrace{\frac{1}{2}(\sqrt{y}+1)}_{\tilde{x}}\underbrace{(1 - \sqrt{y})\,dy}_{dA} = \int_0^1 \frac{1}{2}(1 - y)\,dy$

$$= \frac{1}{2}\left(y - \frac{1}{2}y^2\right)_0^1 = \frac{1}{2}\left(1 - \frac{1}{2}\right) = \boxed{\frac{1}{4} = 0.25\ m^3}$$

TI-89: $\int(.5 * (1 - y), y, 0, 1) = .25\ m^3$

$\bar{x}$ centroid: $\bar{x} = \frac{.25}{.333} = \frac{1/4}{1/3} = \boxed{\frac{3}{4} = .75\ m = \bar{X}}$

$\bar{y}$ **Numerator:** $= \int_0^1 \underset{\tilde{y}}{y}\,\underbrace{(1 - y^{1/2})\,dy}_{dA} = \int_0^1 (y - y^{3/2})\,dy$

$$= \left(\frac{1}{2}y^2 - \frac{2}{5}y^{5/2}\right)_0^1 = \frac{1}{2} - \frac{2}{5} = \frac{5}{10} - \frac{4}{10} = \boxed{\frac{1}{10}\ m^3}$$

TI-89: $\int(y - y\text{^}1.5, y, 0, 1) = 0.10\ m^3$

$\bar{y}$ centroid: $\bar{y} = \frac{1/10}{1/3} = \boxed{\frac{3}{10} = 0.30\ m = \bar{y}}$